

International Islamic University Chittagong (IIUC)

Department of Computer and Communication Engineering

Midterm Examination

Program: B.sc (Engg.)

Course Code: CCE – 3503

Total Marks: 30

Semester: Spring 2026

Course Title: Information Theory & Error Correction Coding

Time: 1 Hours 30 Minutes

(i) Answer all the questions. The figures in the right-hand margin indicate full marks. (ii) Course Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.						
Course Outcomes (COs) of the Questions						
CLO1	Familiarization with the various fields/trends of communications like Information theory, error detection and error correction related to radio and telecommunication.					
CLO2	Analysis different types of error control code properties for secure and error free efficient communications.					
Bloom's Levels of the Questions						
Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

- Q1** a) Consider a discrete memoryless source with alphabet $A=\{s_0, s_1, s_2\}$ with respective probabilities 0.2, 0.2 and 0.6.
 i) Calculate the entropy of the source $H(A)$, in bits/symbol.
 ii) Calculate the entropy of the second-order extension of the source, $H(A^2)$.
 b) Consider the four codes listed below:

CLO2 E 4

CLO2 An 6

Symbol	Code I	Code II	Code III	Code IV
s_0	1	0	0	00
s_1	01	10	01	01
s_2	001	110	11	10
s_3	0001	111	110	11

- i) Two of these four codes are prefix codes. Identify them, and construct their individual decision trees.
 ii) "Explain the significance of the Kraft-McMillan Inequality in relation to these codes. Specifically, address the statement: "A code may satisfy the Kraft-McMillan Inequality but still fail to be a prefix code." Provide a brief justification using one of the non-prefix codes from the table above as an example.

- Q2.** a) Find the source-codes for the following binary sequence using *Lempel-Ziv algorithm* and on the assumption that the code word dictionary already contains 0 and 1 at the start-up.

CLO2 E 4

0010110010100110

- b) Consider the markov chain with three states $S=\{1,2,3\}$ that has the following transition matrix CLO2 E 6

$$P = \begin{bmatrix} .4 & .4 & .2 \\ .3 & .5 & .2 \\ .1 & .6 & .3 \end{bmatrix}$$

- Draw the state transition diagram for this chain.
- If we know $P(X_1=1)=0.5$ and $P(X_1=2)=0.3$, find $P(X_1=3, X_2=1, X_3=3)$.

- Q3. A Discrete Memoryless Source has an alphabet of six symbols whose probabilities of occurrence are as described here: CLO2 E 4

Symbol	Probability
s_0	0.35
s_1	0.25
s_2	0.15
s_3	0.10
s_4	0.10
s_5	0.05

- Use the Huffman encoding procedure to determine a binary code for the source output.
- Determine the average number of binary digits per source symbol.
- Determine the coding efficiency of your Huffman Code.

OR

- Q3. A discrete memoryless source has an alphabet of six symbols whose probabilities of occurrence are as described here: CLO2 E 4

Symbol	Probability
s_0	0.4
s_1	0.2
s_2	0.1
s_3	0.1
s_4	0.1
s_5	0.1

- Compute the Huffman code for this source.
- Find the average code word length and entropy of this code.
- Determine the coding efficiency of your Huffman Code.

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(ii) Course Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.

Course Outcomes (COs) of the Questions

CLO1	Familiarization with the various fields/trends of communications like Information theory, error detection and error correction related to radio and telecommunication.
CLO2	Analysis different types of error control code properties for secure and error free efficient communications.

Bloom's Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

1. a) Define: *Entropy*. For a DMS X with two symbols x_1 and x_2 and $P\{x_1\} = 0.9$ and $P\{x_2\} = 0.1$. Determine second extension of the source. Show that $H(X^2) = 2H(X)$. CLO2 E 4

- b) Consider the four codes listed below: CLO2 An 6

Symbol	Code I	Code II	Code III	Code IV
s_0	0	0	0	00
s_1	10	01	01	01
s_2	110	001	011	11
s_3	1110	0010	110	1000
s_4	1111	0011	111	1001

- i) Two of these four codes are prefix codes. Identify them, and construct their individual decision trees.
- ii) "Meeting Kraft-McMillan Inequality does not give the guarantee of being a prefix code"-justify the statement considering all four codes.

- Q2. a) Consider a Markov chain with three states $S = \{1, 2, 3\}$ and the following transition matrix:

CLO2 E 6

$$P = \begin{bmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{4} & \frac{3}{4} & 0 \end{bmatrix}$$

$$\ln \frac{4}{3} \approx \frac{1}{12}$$

- b) If the initial distribution is $P(X_1=1)=1/3$, $P(X_1=2)=1/6$, $P(X_1=3)=1/2$, find $P(X_1=3, X_2=1, X_3=2)$.

Find the source codes for the

$$P(x_{122})$$

- b) Find the source codes for the following sequence using the Lempel-Ziv (LZ78) compression algorithm, assuming that the initial dictionary already contains the symbols 'A' and 'B'.

CLO2 E 4

Given sequence:

ABA|ABA|BABBABA|BABABBABBA|ABBABABA|B

- Q3.** In a communication system, the source transmits five different messages say x_1, x_2, x_3, x_4, x_5 with the probabilities of 0.3, 0.15, 0.25, 0.05, 0.25 respectively. If the number of symbols used to code these messages is 2, then find the code word for each message and the coding efficiency by applying Huffman Coding method.

CL02 E 10

OR

- Q3.** A discrete memoryless source has an alphabet of six symbols $x_1, x_2, x_3, x_4, x_5, x_6$ whose probabilities of occurrence are as 0.4, 0.2, 0.1, 0.1, 0.1, 0.1;

CLO2 E 10

- (i) Compute the Huffman code for this source.
- (ii) Find the average code word length and entropy of this code.
- (iii) On what percentage the average code word length does exceed the entropy.

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Program: **B.sc (Engg.)**
 Course Code: **CCE – 3503**
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Semester: **Spring 2025**
 Course Title: **Information Theory & Error Correction Coding**
 Time: **1 Hour 30 Minutes**

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
 (ii) Course Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.

Course Outcomes (COs) of the Questions

- CLO1** Familiarization with the various fields/trends of communications like Information theory, error detection and error correction related to radio and telecommunication.
CLO2 Analysis different types of error control code properties for secure and error free efficient communications.

Bloom's Levels of the Questions

Letter Symbols Meaning	R	U	Ap	An	E	C
	Remember	Understand	Apply	Analyze	Evaluate	Create

- Q1. a)** Consider a discrete memoryless source with alphabet $\zeta = \{s_0, s_1, s_2\}$ with respective probabilities 0.25, 0.25 and 0.5. Find $H(\zeta)$ and $H(\zeta^2)$ and establish a relation between $H(\zeta)$ and $H(\zeta^2)$. **CLO2 E 4**
- b)** Compute the joint entropy by considering the following joint probability of a vowel and a consonant occurring together in the same syllable. **CLO1 An 6**

f(x, y)	p	t	k	f(y)
a	$\frac{1}{16}$	$\frac{3}{8}$	$\frac{1}{16}$	$\frac{1}{2}$
i	$\frac{1}{16}$	$\frac{3}{16}$	0	$\frac{1}{4}$
u	0	$\frac{3}{16}$	$\frac{1}{16}$	$\frac{1}{4}$
f(x)	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{1}{8}$	

- Q2. a)** Find the source-codes for the following binary sequence using *Lempel-Ziv algorithm* and on the assumption that the code word dictionary already contains 0 and 1 at the start-up. **CLO2 E 4**
- 1 1 1 0 1 0 0 1 1 0 0 0 1 0 1 1 0 1 0 0
- b)** Consider the markov chain with three states $S = \{1, 2, 3\}$ that has the following transition matrix **CLO2 E 6**

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

- i) Draw the state transition diagram for this chain.
 ii) If we know $P(X_1=1) = P(X_1=2) = 1/4$, find $P(X_1=3, X_2=2, X_3=1)$.

Q3.

A Discrete Memoryless Source has an alphabet of eight symbols whose probabilities of occurrence are as described here:

CLO2 E 4
+
3
+
3

Symbol	Probability
s_0	0.25
s_1	0.20
s_2	0.15
s_3	0.12
s_4	0.10
s_5	0.08
s_6	0.05
s_7	0.05

- Use the Huffman encoding procedure to determine a binary code for the source output.
- Determine the average number of binary digits per source symbol.
- Determine the entropy of the source and compare it with average number.

OR

Q3.

A discrete memoryless source has an alphabet of six symbols whose probabilities of occurrence are as described here:

CLO2 E 4
+
3
+
3

Symbol	Probability
s_0	0.4
s_1	0.2
s_2	0.1
s_3	0.1
s_4	0.1
s_5	0.1

- Compute the Huffman code for this source.
- Find the average code word length and entropy of this code.
- On what percentage the average code word length does exceed the entropy.

G_c, T_c
 $S = \{s_0, s_1, s_2\}$
 Probabilities: 0.25, 0.25, 0.5.

$$\text{Entropy; } H = \sum P_k \log_2 P_k$$

now,

Symbol	Probabilities (P_k)
s_0	0.25
s_1	0.25
s_2	0.5

$$= -[0.25 \log_2 0.25 + 0.25 \log_2 0.25 + 0.5 \log_2 0.5]$$

$$= 1.5 \text{ Bits/msg.}$$

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Midterm Examination

Program: B.sc (Engg.)

Course Code: CCE – 3503

Total Marks: 15

Semester: Autumn 2024

Course Title: Information Theory & Error Correction Coding

Time: 30 Minutes

(i) Answer all the questions. The figures in the right-hand margin indicate full marks.						
(ii) Course Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.						
Course Outcomes (COs) of the Questions						
CLO1	Familiarization with the various fields/trends of communications like Information theory, error detection and error correction related to radio and telecommunication.					
CLO2	Analysis different types of error control code properties for secure and error free efficient communications.					
Bloom's Levels of the Questions						
Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Q1.	a)	Consider the three codes listed below:					CLO2	An	10
		<i>Symbol</i>	<i>Code I</i>	<i>Code II</i>	<i>Code III</i>	<i>Code IV</i>			
		<i>s₀</i>	0	0	0	00			
		<i>s₁</i>	10	01	01	01			
		<i>s₂</i>	110	001	011	10			
		<i>s₃</i>	1110	0010	110	110			
		<i>s₄</i>	1111	0011	111	111			
		i) Identify which the prefix code from above table and construct its decision trees; ii) "Meeting Kraft-McMillan Inequality does not give the guarantee of being a prefix code"-justify the statement considering all three codes.							

Q2.	A Discrete Memoryless Source has an alphabet of five symbols whose probabilities of occurrence are as described here:	CLO2	E	5																
	<table border="1"><thead><tr><th>Symbol</th><th>Probability</th><th>Code</th></tr></thead><tbody><tr><td>s_0</td><td>0.40</td><td>00</td></tr><tr><td>s_1</td><td>0.20</td><td>10</td></tr><tr><td>s_2</td><td>0.20</td><td>11</td></tr><tr><td>s_3</td><td>0.10</td><td>010</td></tr><tr><td>s_4</td><td>0.10</td><td>011</td></tr></tbody></table> 2.2 (i) Determine the average number of binary digits per source symbol. (ii) Determine the entropy of the source and compare it with average number (Efficiency).	Symbol	Probability	Code	s_0	0.40	00	s_1	0.20	10	s_2	0.20	11	s_3	0.10	010	s_4	0.10	011	
Symbol	Probability	Code																		
s_0	0.40	00																		
s_1	0.20	10																		
s_2	0.20	11																		
s_3	0.10	010																		
s_4	0.10	011																		

.OR

Q2.	A Discrete Memoryless Source has an alphabet of five symbols whose probabilities of occurrence are as described here: <table border="1"><thead><tr><th>Symbol</th><th>Probability</th><th>Code</th></tr></thead><tbody><tr><td>s_0</td><td>0.30</td><td>00</td></tr><tr><td>s_1</td><td>0.25</td><td>01</td></tr><tr><td>s_2</td><td>0.25</td><td>10</td></tr><tr><td>s_3</td><td>0.15</td><td>110</td></tr><tr><td>s_4</td><td>0.05</td><td>111</td></tr></tbody></table> (i) Determine the average number of binary digits per source symbol. (ii) Determine the entropy of the source and compare it with average number (Efficiency).	Symbol	Probability	Code	s_0	0.30	00	s_1	0.25	01	s_2	0.25	10	s_3	0.15	110	s_4	0.05	111	CLO2	E	5
Symbol	Probability	Code																				
s_0	0.30	00																				
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CLO1	Familiarization with the various fields/trends of communications like Information theory, error detection and error correction related to radio and telecommunication.
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Bloom's Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Q1	a)	What is information? Why logarithmic function is used to define information?	CLO2	E	4																													
	b)	Consider the four codes listed below: <table><thead><tr><th>Symbol</th><th>Code I</th><th>Code II</th><th>Code III</th><th>Code IV</th></tr></thead><tbody><tr><td>S₀</td><td>0</td><td>0</td><td>0</td><td>00</td></tr><tr><td>S₁</td><td>10</td><td>01</td><td>01</td><td>01</td></tr><tr><td>S₂</td><td>110</td><td>001</td><td>011</td><td>10</td></tr><tr><td>S₃</td><td>1110</td><td>0010</td><td>110</td><td>110</td></tr><tr><td>S₄</td><td>1111</td><td>0011</td><td>111</td><td>111</td></tr></tbody></table> (i) Two of these four codes are prefix codes. Identify them, and construct their individual decision trees. (ii) "Meeting Kraft-Inequality does not give the guarantee of being a prefix code"-justify the statement considering all four codes.	Symbol	Code I	Code II	Code III	Code IV	S ₀	0	0	0	00	S ₁	10	01	01	01	S ₂	110	001	011	10	S ₃	1110	0010	110	110	S ₄	1111	0011	111	111	CLO2	An
Symbol	Code I	Code II	Code III	Code IV																														
S ₀	0	0	0	00																														
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S ₂	110	001	011	10																														
S ₃	1110	0010	110	110																														
S ₄	1111	0011	111	111																														
Q2	a)	Use the Lempel-Ziv algorithm to encode the following sequence. Assume that the binary symbols 0 and 1 are already in the codebook. 11101001100010110100	CLO2	E	4																													

	b) Consider the Markov chain with three states, $S=\{1,2,3\}$ that has the following transition matrix $\begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$ (i) Draw the state transition diagram for this chain. (ii) If $P(X_1=1)=P(X_1=2)=\frac{1}{4}$, find out $P(X_1=3, X_2=2, X_3=1)$	CLO2	E	6																		
Q3	A Discrete Memoryless Source has an alphabet of eight symbols whose probabilities of occurrence are as described here: <table border="1"> <thead> <tr> <th>Symbol</th><th>Probability</th></tr> </thead> <tbody> <tr> <td>S_0</td><td>0.25</td></tr> <tr> <td>S_1</td><td>0.20</td></tr> <tr> <td>S_2</td><td>0.15</td></tr> <tr> <td>S_3</td><td>0.12</td></tr> <tr> <td>S_4</td><td>0.10</td></tr> <tr> <td>S_5</td><td>0.08</td></tr> <tr> <td>S_6</td><td>0.05</td></tr> <tr> <td>S_7</td><td>0.05</td></tr> </tbody> </table> (i) Use the Huffman encoding procedure to determine code words for the source output. (ii) Determine the average code word length per source symbol. (iii) Determine the entropy of the source and compare it with average codeword length. (efficiency)	Symbol	Probability	S_0	0.25	S_1	0.20	S_2	0.15	S_3	0.12	S_4	0.10	S_5	0.08	S_6	0.05	S_7	0.05	CLO2	E	4 + 3 + 3
Symbol	Probability																					
S_0	0.25																					
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S_3	0.12																					
S_4	0.10																					
S_5	0.08																					
S_6	0.05																					
S_7	0.05																					
OR																						
Q3	A discrete memoryless source has an alphabet of six symbols $S_0, S_1, S_2, S_3, S_4, S_5$ whose probabilities of occurrence are 0.4, 0.2, 0.1, 0.1, 0.1, 0.1 respectively (i) Compute the Huffman code for this source. (ii) Find the average code word length and entropy of this code. (iii) On what percentage the average code word length does exceed the entropy.	CLO2	E	4 + 3 + 3																		

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- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

- CLO1** Familiarization with the various fields/trends of communications like Information theory, error detection and error correction related to radio and telecommunication.
CLO2 Analysis different types of error control code properties for secure and error free efficient communications.

Bloom's Levels of the Questions

Letter Symbols
Meaning

R U Ap An E C
Remember Understand Apply Analyze Evaluate Create

1. a) Consider a discrete memoryless source with alphabet $\mathcal{Z} = \{s_0, s_1, s_2\}$ with respective probabilities 0.25, 0.25 and 0.5. Find $H(\mathcal{Z})$ and $H(\mathcal{Z}^2)$ and establish a relation between $H(\mathcal{Z})$ and $H(\mathcal{Z}^2)$. CLO2 E
b) Consider the Markov chain with three states $S = \{1, 2, 3\}$, that has the following transition matrix CLO2 An

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

- a. Draw the state transition diagram for this chain.
b. If we know $P(X_1 = 1) = P(X_1 = 2) = \frac{1}{4}$, find $P(X_1 = 3, X_2 = 2, X_3 = 1)$.

2. a) Consider the four codes listed below: CLO2 E

Symbol	Code I	Code II	Code III	Code IV
s_0	0	0	0	00
s_1	10	01	01	01
s_2	110	001	011	10
s_3	1110	0010	110	110
s_4	1111	0011	111	111

- a. Two of these four codes are prefix codes. Identify them, and construct their individual decision trees.
b. "Meeting **Kraft-McMillan Inequality** does not give the guarantee of being a prefix code"-justify the statement considering all four codes.
b) Find the source-codes for the following binary sequence using **Lempel-Ziv algorithm** and on the assumption that the code word dictionary already contains 0 and 1 at the start-up. CLO2

11101001100010110100



- Q3.** A Discrete Memoryless Source has an alphabet of six symbols whose probabilities of occurrence are as described here: **CLO2 E**

Symbol	Probability
s_0	0.4
s_1	0.2
s_2	0.1
s_3	0.1
s_4	0.1
s_5	0.1

- I. Use the Huffman encoding procedure to determine a binary code for the source output.
- II. Determine the average number of binary digits per source symbol.
- III. Determine the entropy of the source and compare it with average number.

OR

- Q3.** A Discrete Memoryless Source has an alphabet of six symbols whose probabilities of occurrence are as described here: **CLO2**

Symbol	Probability
s_0	0.30
s_1	0.25
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- I. Use the Huffman encoding procedure to determine a binary code for the source output.
- II. Determine the average number of binary digits per source symbol.
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Bloom's Levels of the Questions						
Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Q1.	a)	A zero memory source contains $X=\{x_1, x_2, x_3, x_4\}$ with $P(X)=\{1/2, 1/4, 1/8, 1/8\}$ i) Determine <i>entropy</i> of the source. ii) Determine second extension of the source. Show that $H(X^2)=2H(X)$	CLO2	E	6																								
	b)	Consider the three codes listed below: <table border="1"><thead><tr><th>Source Symbol</th><th>Probability of Occurrence</th><th>Code I</th><th>Code II</th><th>Code III</th></tr></thead><tbody><tr><td>s_0</td><td>0.5</td><td>0</td><td>0</td><td>0</td></tr><tr><td>s_1</td><td>0.25</td><td>1</td><td>10</td><td>01</td></tr><tr><td>s_2</td><td>0.125</td><td>00</td><td>110</td><td>011</td></tr><tr><td>s_3</td><td>0.125</td><td>11</td><td>111</td><td>0111</td></tr></tbody></table> i) Identify which one is the prefix code and construct its decision trees; ii) "Meeting <i>Kraft-McMillan Inequality</i> does not give the guarantee of being a prefix code"-justify the statement considering all three codes.	Source Symbol	Probability of Occurrence	Code I	Code II	Code III	s_0	0.5	0	0	0	s_1	0.25	1	10	01	s_2	0.125	00	110	011	s_3	0.125	11	111	0111	CLO2	An
Source Symbol	Probability of Occurrence	Code I	Code II	Code III																									
s_0	0.5	0	0	0																									
s_1	0.25	1	10	01																									
s_2	0.125	00	110	011																									
s_3	0.125	11	111	0111																									

Q2.

Determine Shannon-Fano code. Calculate efficiency & redundancy of Shannon-Fano coding.

CLO2

E

10

Message	Probability
M_1	1/2
M_2	1/4
M_3	1/8
M_4	1/16
M_5	1/32
M_6	1/64
M_7	1/128
M_8	1/128

Q3.

A Discrete Memoryless Source has an alphabet of eight symbols whose probabilities of occurrence are as described here:

CLO2

E

4+3+3

Symbol	Probability
s_0	0.25
s_1	0.20
s_2	0.15
s_3	0.12
s_4	0.10
s_5	0.08
s_6	0.05
s_7	0.05

- Use the Huffman encoding procedure to determine a binary code for the source output.
- Determine the average number of binary digits per source symbol.
- Determine the entropy of the source and compare it with average number.

OR

Q3.

A Discrete Memoryless Source has an alphabet of eight symbols whose probabilities of occurrence are as described here:

CLO2

E

4+3+3

Symbol	Probability
s_0	0.30
s_1	0.25
s_2	0.20
s_3	0.12
s_4	0.08
s_5	0.05

- (i) Use the Huffman encoding procedure to determine a binary code for the source output.
- (ii) Determine the average number of binary digits per source symbol.
- (iii) Determine the entropy of the source and compare it with average number.